

Employee Salary Details Using an Interface

Java Program

Question

Write a Java program that defines a **Salary** interface and uses it to display salary details for regular and contract employees. Select the employee type from the first character of the employee ID.

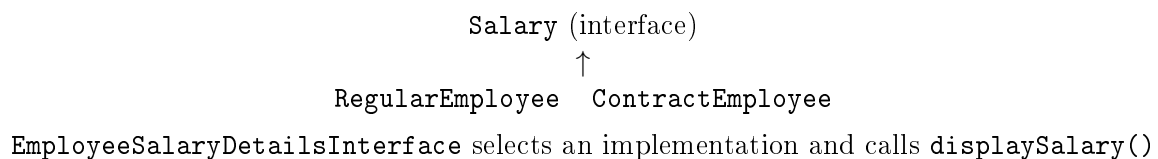
Steps for Execution

1. Save the source code as **EmployeeSalaryDetailsInterface.java**.
2. Open a terminal in the folder containing the file.
3. Compile and run the program using the commands shown below.
4. Enter a non-empty employee ID beginning with **R** for a regular employee or **C** for a contract employee.

Algorithm

1. Define the **Salary** interface with the **displaySalary()** method.
2. Implement the interface in **RegularEmployee** and **ContractEmployee**; each class adds its pay components and prints the total.
3. Read the employee ID from standard input.
4. If its first character is **R** or **r**, create a regular employee; if it is **C** or **c**, create a contract employee.
5. If the first character matches neither type, print an invalid-ID message and stop.
6. Otherwise, call **displaySalary()** on the selected employee.

Class Relationships



Source Code

```
import java.util.Scanner;

// Interface
interface Salary {
    void displaySalary();
}

// Regular Employee
class RegularEmployee implements Salary {
    int basicPay = 25000;
    int hra = 15000;
    int ta = 5000;
```

```

    public void displaySalary() {
        int total = basicPay + hra + ta;

        System.out.println("Salary Details:");
        System.out.println("Basic Pay: " + basicPay);
        System.out.println("HRA: " + hra);
        System.out.println("T.A: " + ta);
        System.out.println("Total Amount: " + total);
    }
}

// Contract Employee
class ContractEmployee implements Salary {
    int basicPay = 12000;
    int hra = 0;
    int ta = 3000;

    public void displaySalary() {
        int total = basicPay + hra + ta;

        System.out.println("Salary Details:");
        System.out.println("Basic Pay: " + basicPay);
        System.out.println("HRA: " + hra);
        System.out.println("T.A: " + ta);
        System.out.println("Total Amount: " + total);
    }
}

// Main class
public class EmployeeSalaryDetailsInterface {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Employee Id: ");
        String id = sc.nextLine();

        Salary employee;

        if (id.charAt(0) == 'R' || id.charAt(0) == 'r') {
            employee = new RegularEmployee();
        }
        else if (id.charAt(0) == 'C' || id.charAt(0) == 'c') {
            employee = new ContractEmployee();
        }
        else {
            System.out.println("Invalid Employee Id");
            sc.close();
            return;
        }

        employee.displaySalary();

        sc.close();
    }
}

```

Compilation and Execution

Compile and run from the directory containing the source file:

```
javac EmployeeSalaryDetailsInterface.java
java EmployeeSalaryDetailsInterface
```

Sample Output

For the input R101:

```
Enter Employee Id: R101
Salary Details:
Basic Pay: 25000
HRA: 15000
T.A: 5000
Total Amount: 45000
```

For a contract employee, the total amount is 15000. An ID beginning with another character prints **Invalid Employee Id**. This version expects a non-empty ID; empty input is not handled before reading its first character.